

María Fernanda Peralta Morales

PhD Candidate in Computational Chemistry · Universitat Autònoma de Barcelona

Profile

Chemist and PhD candidate in Computational Chemistry at the Universitat Autònoma de Barcelona (InSiliChem group). My doctoral research focuses on the rational design of artificial metalloenzymes and *de novo* metallopeptides for biocatalysis and drug design through molecular dynamics, structure prediction and mechanistic modelling.

I also develop open-source Python tools to automate and deepen the analysis of simulations, and I aim to integrate machine learning and AI models into these pipelines.

My background also spans analytical method validation in the pharmaceutical industry, transcriptomics and neuroscience research, teaching and science outreach.

Contact

+34 688 65 84 28

MariaFernanda.Peralta@uab.cat

Barcelona, Spain

linkedin.com/in/maria-fernanda-peralta-morales-55b731221

github.com/insilichem

ORCID: 0009-0006-9723-3135

Languages

Spanish
Native

English
B2 – TOEFL iBT: 92

German
Basic (A2)

Catalan
Basic (A2)

Education

PhD in Chemistry (Official Doctoral Programme)

Universitat Autònoma de Barcelona (UAB) · 09/2025 – present

Provisional thesis title: *Computational Strategies for the Rational Design of Artificial Metalloenzymes and De Novo Metallopeptides*

Supervisors: Dr. Jean-Didier Maréchal & Dr. Giuseppe Sciortino · InSiliChem group

Master's Degree in Bioinformatics

Universitat Autònoma de Barcelona (UAB) · 2024 – 2025

Specialisation: Structure and Function of Proteins and Drug Design

Master's thesis: *Unraveling the Mechanism of Gold-Catalyzed Alkenynone Reaction in a Protein Environment*

Thesis grade: **9.5/10** · Overall grade: **8.40/10**

Bachelor's Degree in Chemistry

Universidad Nacional de Colombia · 2017 – 2022

GPA: **4.2/5.0** · Spanish equivalent: **7.81/10**

Graduating cohort average: 4.0/5.0 (Spanish equivalent: 7.0/10)

Degree thesis: *Evaluation of in vitro neuroprotection and agonist activity of the compounds 5,7-dimethoxy-4H-chromen-4-one and 10H-furo[3,2-a]carbazole extracted from* Zanthoxylum caribaeum.

Diploma Course in HPLC and LC-MS

Universidad Nacional de Colombia · 2023

High Performance Liquid Chromatography and Liquid Chromatography–Mass Spectrometry.

Research Experience

PhD Researcher – InSiliChem Group (Computational Chemistry)

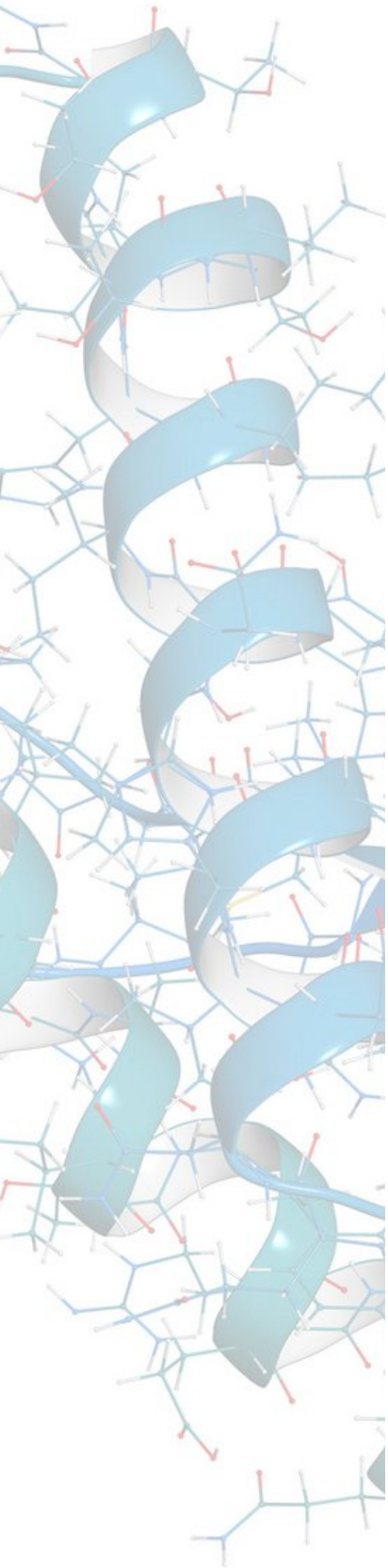
Universitat Autònoma de Barcelona · 09/2025 – present · Barcelona, Spain

- Rational design of artificial metalloenzymes for asymmetric biocatalysis, including mechanistic modelling of Au-catalysed alkenynone cyclisation.
- Design and characterisation of *de novo* metallopeptides for biocatalysis and drug design.
- Molecular dynamics simulations (AMBER) and structure prediction (AlphaFold) to study protein structure–function relationships.
- Development of Python tools and automated workflows for trajectory analysis on GNU/Linux, using AI-assisted coding.

Research Assistant – Transcriptomics Group (Bioinformatics and Systems Biology)

Universidad Nacional de Colombia · 2022 – 2024 · Bogotá, Colombia

- Computational research in protein design to enhance catalytic activity.
- Analysis of genomic data (microarray, RNA-seq, single-cell RNA-seq) and construction of gene networks.



María Fernanda Peralta Morales

Research Experience (cont.)

Undergraduate Researcher – Alzheimer Group (Neurosciences and Cell Death)
Universidad Nacional de Colombia · **2021 – 2022** · Bogotá, Colombia

- Evaluation of the neuroprotective and agonist activity of natural compounds from *Zanthoxylum caribaeum* using cell viability (MTT) and luciferase reporter (LXRE-LUC) assays.

Research Software

Gephyra – Water-bridge network analysis for molecular dynamics · 2026
github.com/insilichem/Gephyra · Peralta-Morales, M. F., Sciortino, G. & Maréchal, J.-D.

- Pip-installable Python package with a command-line interface to discover, track and statistically analyse water-mediated hydrogen-bond networks in MD trajectories.
- Probabilistic hydrogen-bond scoring, weighted graph path search (NetworkX) and Fréchet-distance hierarchical clustering of pathways; exports visualisation scripts for PyMOL, VMD and UCSF Chimera.

Water Counting – Water molecules per plane in UCSF Chimera · 2026
github.com/insilichem/Water-count-per-plane · Peralta-Morales, M. F., Sciortino, G. & Maréchal, J.-D.

- Python script for UCSF Chimera that counts water molecules across defined planes of a molecular system.

Conference Contributions

POSTER **2026** · ESPA 2026 – Electronic Structure Principles and Applications & IV Biennial Meeting of the RSEQ Group in Chemistry and Computation · Salamanca, Spain · 30 June – 3 July 2026
Modelling Au-based Asymmetric Catalysis in Artificial Metalloenzymes.

ORAL **2026** · 14th Trobada de Joves Investigadors dels Països Catalans · Lleida, Spain
Modelling Au-Based Asymmetric Catalysis in Artificial Metalloenzymes: Mechanistic Insights into Gold-Catalyzed Alkenynone Cyclization.

POSTER **2021** · CLAQ – Latin American Congress of Chemistry · Instituto Nacional de Metrología, Colombia
Review of the method for determining pH in emulsion-type water-based paints in Colombia: metrological update.

Professional Experience

HPLC Stability and Validation Analyst
AQM SAS (pharmaceutical sector) · **2022 – 2023** · Bogotá, Colombia

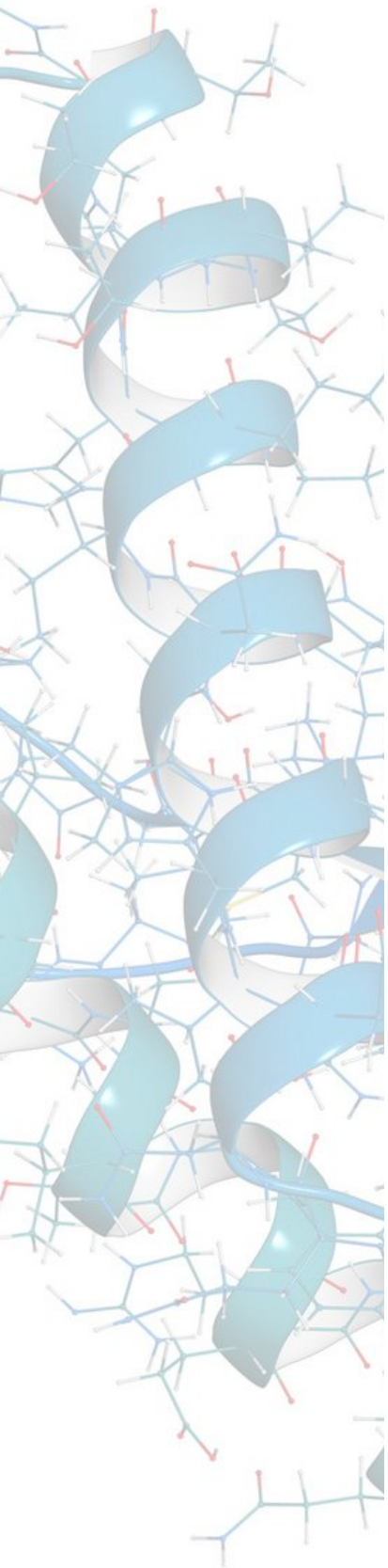
- Stability studies and analytical method validation/verification by HPLC for pharmaceutical products.
- Data analysis and problem-solving within a multidisciplinary team (5+ people).

Organic and Physical Chemistry Laboratory Assistant
Universidad Nacional de Colombia · **2021 – 2022** · Bogotá, Colombia

- Supported undergraduate laboratory sessions in organic and physical chemistry.

Science Teacher
RENFORT · **2017 – 2019** · Bogotá, Colombia

- Taught science to school students.



María Fernanda Peralta Morales

Skills

Computational skills

Molecular dynamics simulations
System preparation, simulation and analysis of protein and metalloprotein systems with AMBER · Parametrisation of metal centres with MCPB.py (bonded model). Trajectory analysis of structural stability, conformational change, hydrogen bonding and solvent/water networks (MDAnalysis, VMD, UCSF Chimera, ChimeraX, PyMOL).

Mechanistic and quantum chemical modelling
DFT calculations with Gaussian 16 on cluster models of the metal coordination site · B3LYP-D3 (GD3 empirical dispersion), SDD for the metal and 6-31G(d,p) for all other atoms · Reaction mechanism modelling of Au-catalysed alkenynone cyclisation in protein environments.

Protein structure modelling and design
Structure prediction and homology modelling (AlphaFold, Modeller) · Rational design of artificial metalloenzymes and *de novo* metallopeptides · Structure–function analysis and molecular visualisation (PyMOL, UCSF Chimera, ChimeraX, Avogadro) · Introductory experience with Rosetta and RFdiffusion.

Docking and ligand-based modelling
Protein–ligand docking (GOLD, AutoDock) · Metalloligand–protein docking in GOLD with metal-compatible parameters: activation of vacant coordination sites with dummy atoms, modelling of metal–residue coordination bonds through the GoldScore scoring function, and side-chain flexibility via rotamer libraries · Pharmacophore modelling (LigandScout).

Scientific programming and software development
Python for scientific computing (NumPy, SciPy, MDAnalysis, NetworkX) · Development and packaging of open-source command-line tools · UCSF Chimera scripting (Python 2.7) · Workflow automation with Bash on GNU/Linux · Job submission and management on HPC clusters via SSH (SLURM workload manager) · Conda environments · Git/GitHub · R/RStudio · AI-assisted software development (Claude, Gemini, Jules).

Data analysis
Graph and network analysis · Hierarchical clustering and trajectory similarity metrics (Fréchet distance) · Statistical analysis · Genomic data (microarray, RNA-seq, single-cell RNA-seq, gene network construction).

Experimental skills

Chemical analysis
Chromatography and mass spectrometry: HPLC (stability studies, method validation), LC-MS. Structural elucidation: IR spectroscopy, NMR. Quantification: potentiometric titration, UV-Vis spectrophotometry, Kjeldahl nitrogen determination.

Biochemistry
Cell culture: cell viability (MTT) and luciferase reporter (LXRE-LUC) assays.

Science Outreach and Volunteering

Communications Committee Leader – American Chemical Society Student Chapter
Universidad Nacional de Colombia · **2019 – 2021**

- Led science communication activities for children and young people across Colombia.

Camp Counselor – CCUSA
Summers 2016 – 2021 · New Hampshire and Arizona, USA

- Coordinated educational activities for children in an English-speaking international environment.

References

Dr. Jean-Didier Maréchal – PhD supervisor
InSiliChem group, Department of Chemistry, Universitat Autònoma de Barcelona · JeanDidier.Marechal@uab.cat

Dr. Adrián Gabriel Sandoval Hernández, Full Professor (Profesor Titular)
Chemist, MSc, PhD in Science · Department of Chemistry, Universidad Nacional de Colombia · agsandovalh@unal.edu.co